Chemical labels for our molecule catalogue

A short guide for Tiansi & Pietro August 25, 2026

1. The goal (in plain words)

We have a catalogue of about **1.95 million commercial "building-block" molecules** (small chemical parts used to assemble bigger molecules). Our downstream AI generator needs a reliable *category label* for each building block — like tagging every item in a huge warehouse with the correct department, aisle and shelf. The generator uses these tags as **rules**, so the tags must be **correct, not guessed**. The official tagging system we use is called **ChemOnt / ClassyFire**.

Two concrete goals: **(1)** collect a large set of *genuine* (officially computed) tags for our building blocks; **(2)** check whether a fast in-house "auto-tagger" can reproduce those tags well enough to one day tag all 1.95M molecules automatically.

2. How the chemical families are structured (important)

The families are organised as a **strict tree** — a rooted hierarchy — *not* a free-form network or web. Concretely:

- Every category has **exactly one parent**, and every category traces back to a **single root** called "Chemical entities". We verified this on the real file: **4,825 categories**, one root, **zero** categories with two parents, no loops, up to **11 levels** deep.
- So a molecule's classification is a **single path from the root down to a specific family**, exactly like *Department* → *Aisle* → *Shelf*. Broad levels have names like *kingdom, superclass, class, subclass*.

> **Real example** (one of our building blocks). Its path down the tree:
>
> `Chemical entities` → `Organic compounds` → `Benzenoids` → `Benzene and substituted derivatives` → `Benzonitriles`

Small nuance: the tree above is the fixed backbone. On top of it, ClassyFire may also list a few "related families" for a molecule as extra context (so a molecule can point to more than one family), but the backbone itself stays a clean tree. Our auto-tagger mirrors this: it picks one primary family (a leaf of the tree) plus a few alternatives.

3. What we did (methods)

- **Fixed reference.** We used the complete official ChemOnt tree, unchanged, as the ground-truth backbone. We store it as two simple tables: the *categories* (nodes) and the *parent→child links* (edges) that make up the tree.
- **Genuine tags only.** We downloaded real tags from two official public services, one molecule at a time, *politely* (respecting their speed limits), fully automatic and restart-safe. The collection ran for about two weeks. We never invented a tag.
- **An honest test.** We split the tagged molecules by their chemical "skeleton" so the test set contains molecule types the auto-tagger never saw during development — no overlap between what it learns from and what it is judged on.

4. What we found (results)

Collected **200,001 genuine tags** (target reached), spanning **26 broad families**, **425 categories** and **1,977 fine categories**. This large, trustworthy dataset is the hard part, and it is done. On a held-out test of **39,021 unseen molecules**, the fast auto-tagger scores:

	auto-tagger (test)	our quality bar	good enough?
Broad family correct	72%	90%	not yet
Category correct	60%	75%	not yet
Exact fine tag correct	13%	60%	not yet

What it means & next steps. The genuine dataset is ready and reliable. The fast auto-tagger is a useful draft but **not accurate enough yet** to tag all 1.95M molecules automatically: doing so now would feed some wrong rules to the generator. So we deliberately **stop before mass auto-tagging** and wait for a human go-ahead. **Next:** use the 200k genuine tags to improve the auto-tagger, then re-test against the same honest benchmark; scale up only once it clears the quality bar.

5. Where to look on GitHub

Repository: `github.com/tacc-code-tacclab/chemont-classyfire-buildingblocks`

- `scripts/v4/` — the main code (data collection, the ChemOnt tree parser, the benchmark, the finalisation).
- `reports/v4/` — all reports and result numbers (this document is `mini_report_tiansi_pietro.pdf`).
- `database/v4/` — the ChemOnt tree itself as two tables: `chemont_nodes.tsv` (the 4,825 categories) and `chemont_edges.tsv` (the parent→child links).
- `src/`, `scripts/`, `tests/` — supporting code and checks.

Note: the raw ZINC structures and the genuine ClassyFire labels are **not** on GitHub — they are large (multi-GB) and their licences restrict redistribution. They live on the lab server under `data/v4_classyfire_groundtruth/` (genuine tags, train/validation/test splits) and `database/v4/dag_v4.db`. GitHub holds the code, the reports, and the tree structure so you can review exactly how everything was done.